Proposed Manuscript Insertions VII
Source-Complete Binary Contacts and the Pseudo-Trace Atlas

Research draft for the marked-width MLC program

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Abstract

For the actual binary f^q pullback system at the first direct-decoration scale, we recover the missing one-step boundary contacts from the source 2^n-component attachment theorem, compute all deeper contacts by a finite normalized recursion, retain endpoint flags and cyclic sides, and construct a one-to-many trace-pullback atlas on the source-native pseudo-puzzle boundary. The atlas is exactly natural under normalization, safe-gap carrier enlargement, and certified finite overlay refinement. The results close the binary contact/trace construction for a fixed source block and finite marked overlay; they do not prove a universal response/Value quotient, MBTT, rigidity, or MLC.

1 Exact binary source package

Let

$$P = Y^{(n-1)q+1}, \qquad g = f^q,$$

and let $Q_L, Q_R \subset P$ be the two actual univalent f^q pullbacks. Write

$$g_a = g|_{Q_a} : Q_a \to P, \qquad \phi_a = g_a^{-1}, \qquad a \in \mathcal{A} = \{L, R\}.$$

For $w = a_1 \cdots a_k$, put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_k}, \quad D_w = \phi_w(P), \quad h_w = g^k|_{D_w} : D_w \to P.$$

The already established closed-piece separation gives $Q_L \cap Q_R = \varnothing$.

The source proves that the domain of g^n has 2^n components. For $s \in \mathcal{A}^n$, let

$$v_s = \phi_s(\alpha).$$

These points are the vertices of P, and the component D_s is attached to the two boundary rays of P landing at v_s.

2 One-step contact recovery

Define

$$C_a = \partial P \cap \partial Q_a, \qquad \mathcal{V}_a = \{v_{at} : t \in \mathcal{A}^{n-1}\}.$$

Theorem 2.1 (Source recovery of the binary contacts)**.** *For $a \in \{L, R\}$:*

1. *the landing vertices of C_a are exactly $\mathcal{V}_a$;*

2. *C_a consists of the maximal common subsegments of the two boundary rays of P landing at those vertices, together with their endpoint flags;*
3. *no nondegenerate equipotential arc belongs to C_a;*
4. *the complete contact is computed by exact matching of the finite typed boundary words of P and Q_a.*

After finite subdivision, g_a maps every open contact atom homeomorphically onto one root-boundary atom.

Proof. For $s = at$, the source component D_s is contained in Q_a and shares the pair of boundary rays of P landing at v_s. A point of such a common ray cannot be interior to $Q_a \subset P$, hence the common ray is also on ∂Q_a. Thus every v_{at} occurs.

Conversely, let an open contact ray of P and Q_a land at a vertex v of P. The source attachment theorem supplies the unique depth-n component D_s attached at v. Its first-level cylinder contains v. Since the two closed first-level cylinders are disjoint and $v \in Q_a$, the first symbol of s must be a. Hence $v = v_{at}$.

The parent and child equipotential levels differ by q, so their open arcs are disjoint. Distinct external rays are disjoint outside the filled Julia set. Thus exact ray-angle matching and intersection of the potential intervals recovers all open contacts. Subdivision at their endpoints and images makes the branch map simplicial. □

Definition 2.2 (Endpoint flag)**.** A boundary flag is a vertex, an incident open boundary atom, the side of the puzzle piece, and the induced cyclic orientation. Bare landing points are not used as endpoint records.

Each g_a induces a partial flag map

$$\sigma_a : \mathscr{F}(C_a) \longrightarrow \mathscr{F}(\partial P)$$

that preserves cyclic order.

3 Complete all-depth contacts

For a word w, set

$$C_w = \partial P \cap \partial D_w, \qquad \widehat{C}_w = h_w(C_w) \subset \partial P, \qquad \widehat{C}_\varnothing = \partial P.$$

Theorem 3.1 (Exact contact recursion)**.** *For every word w and symbol a,*

$$\boxed{\widehat{C}_{wa} = g_a(\widehat{C}_w \cap C_a).}$$

Proof. If $z \in C_{wa}$, nesting $D_{wa} \subset D_w \subset P$ implies $z \in \partial D_w$: otherwise z would be interior to $D_w \subset P$ while lying on ∂P. Hence $x = h_w(z) \in \widehat{C}_w$. Since $D_{wa} = \phi_w(Q_a)$, we also have $x \in \partial Q_a$, and $h_{wa}(z) = g_a(x)$.

Conversely, if $x \in \widehat{C}_w \cap C_a$, then $z = h_w^{-1}(x)$ lies on ∂P by the definition of $\widehat{C}_w$, and on ∂D_{wa} because $x \in \partial Q_a$. Thus $z \in C_{wa}$. □

Theorem 3.2 (Complete binary cylinder contacts)**.** *For finite words u, v:*

1. *if u, v are incomparable, then $D_u \cap D_v = \varnothing$;*
2. *if $v = ut$, then*

$$\boxed{\partial D_u \cap \partial D_v = \phi_u(C_t).}$$

Consequently the two one-step contact graphs determine every all-depth contact.

Proof. At the first differing symbol, incomparable pieces lie in the inverse images of the disjoint closed pieces Q_L, Q_R. If $v = ut$, then $D_v = \phi_u(D_t)$ and $D_u = \phi_u(P)$; apply the closed-piece homeomorphism ϕ_u. □

After subdivision, the normalized contact dynamics is the finite automaton

$$T_a(S) = g_a(S \cap C_a)$$

on subcomplexes of one finite root boundary graph. This is a per-block finite state statement, not a universal bound in n or q.

4 Source endpoint itineraries

For a prefix $w \in \mathcal{A}^k$, $0 \le k \le n$, define

$$\mathcal{V}(w) = \{v_{wt} : t \in \mathcal{A}^{n-k}\}.$$

Theorem 4.1 (Prefix endpoint itinerary)**.** *The landing vertices of the open ray contacts in $\partial P \cap \partial D_w$ are exactly $\mathcal{V}(w)$. Moreover*

$$\boxed{h_w(v_{wt}) = v_t.}$$

The incident boundary-ray flags and their cyclic order are transported by the same shift.

Proof. Every depth-n component $D_{wt} \subset D_w$ is source-attached to the boundary rays at v_{wt}; nesting forces those common rays to remain on the boundary of each intermediate piece. Conversely, a contact vertex of D_w has a unique source depth-n attached component. Its word must begin with w, since otherwise it lies in an incomparable disjoint cylinder. Finally, g^k cancels the first k inverse branches. □

Theorem 4.2 (Post-transient synchronization)**.** *Let $\mathscr{R}_\alpha \subset \partial P$ be the complete pair of root boundary-ray atoms landing at α, with endpoint flags. For every $s \in \mathcal{A}^n$,*

$$\widehat{C}_s = \mathscr{R}_\alpha.$$

The left branch fixes this contact, while the right branch kills it:

$$T_L(\mathscr{R}_\alpha) = \mathscr{R}_\alpha, \qquad T_R(\mathscr{R}_\alpha) = \varnothing.$$

Hence for every continuation t, the contact persists exactly when $t = L^m$; after the first R it is empty forever. The same holds for endpoint flags.

Proof. The prefix theorem at $k = n$ gives the unique normalized landing vertex α for every depth-n word, and h_s maps the attached deeper ray pair onto the complete root ray pair. The left inverse branch is attached at the fixed point α and f^q fixes its ray pair. The right first-level piece is a disjoint closed piece attached at α', so exact boundary-word matching gives no ray atom or endpoint flag in common with $\mathscr{R}_\alpha$. Apply the contact recursion; the empty state is absorbing. □

Corollary 4.3. *The 2^n source fan is transient contact data. Beyond source depth n, one address has only two possible normalized contact states: the fixed α-port or no root contact. This does not bound the number of transient proof-used ports across all words.*

5 One-to-many pseudo-trace atlas

The source pseudo-functor associates a pseudo-piece $\mathbf{Y}$ to each puzzle piece. Its ideal boundary has finitely many O-boundary and J-boundary arcs; at a vertex v the source port is $\partial_v \mathbf{Y}$. A degree-one puzzle branch induces a degree-one covering of pseudo-pieces and extends continuously to the ideal boundary.

Choose a finite root overlay $\mathscr{O}$ containing all contact endpoints, source vertex-port endpoints, and the current finite protected, obstacle and electrode endpoints. Pull it back to $\mathbf{Q}_L, \mathbf{Q}_R$ and refine the root by all images. Let $\mathscr{A}_P$ and $\mathscr{A}_{Q_a}$ be the resulting open atoms. Each child atom K maps homeomorphically to one parent atom $p_a(K)$.

For $J \in \mathscr{A}_P$, put

$$\operatorname{Lift}(J) = \{(a, K) : p_a(K) = J\}.$$

Because each branch has degree one,

$$|\operatorname{Lift}(J)| \le 2.$$

For a pseudo-piece $\mathbf{Y}$ and a finite atom union B, let

$$\mathcal{T}_{\mathbf{Y}}(B) = H^1_{\mathrm{fin}}(\mathbf{Y})/\{u : \operatorname{Tr}_B u = 0\}$$

with its minimum-energy quotient norm. Define the compatible tuple space over J to be the image of the joint branch pullback.

Theorem 5.1 (Binary one-to-many trace atlas)**.** *For every root atom J, actual trace pullback defines a continuous map*

$$\mathcal{R}_J : \mathcal{T}_{\mathbf{P}}(J) \longrightarrow \mathcal{T}_J^{\mathrm{comp}},$$
$$\mathcal{R}_J u = \left((u|_{\mathbf{g}_a(K)}) \circ \mathbf{g}_a|_K\right)_{(a,K)\in \operatorname{Lift}(J)}.$$

It retains exact branch pullback, endpoint flags, side and cyclic order; it is conformally natural for Dirichlet energy and assumes no omitted Euclidean boundary welding.

Proof. The source-induced pseudo-map is conformal and extends to the ideal boundary. Every refined child atom maps homeomorphically to its parent atom. Therefore trace pullback is defined on the intrinsic minimum-energy trace quotient. The compatible tuple space is the image of the joint map, so vertex compatibility is automatic. Conformal invariance gives energy naturality. □

For a word w, direct pullback by the induced $\mathbf{h}_w$ equals the composition of the one-step maps, with canonical reassociation of the compatible tuple spaces. The analogous flag and cyclic-order identities also hold.

6 Normalized-overlay naturality

At a word w, put

$$\mathscr{O}_w = \mathbf{h}_w^{-1}(\mathscr{O}).$$

On $\mathbf{D}_{wa}$,

$$\boxed{\mathbf{h}_{wa} = \mathbf{g}_a \circ \mathbf{h}_w.}$$

Theorem 6.1 (Normalized-overlay naturality)**.** *For every w, a, this identity gives commuting diagrams for:*

1. *contact subgraphs;*

2. *endpoint flags and cyclic order;*
3. *one-to-many trace restrictions;*
4. *compatible trace spaces;*
5. *all finite response forms obtained by minimizing internal traces.*

Two prefixes with the same normalized contact state and live marked overlay therefore have the same future contact/trace behavior.

Proof. The polynomial identity is exact on the selected pieces. The pseudo-functor preserves it. The contact square is the contact-recursion theorem; the flag and trace squares are the boundary extension and functoriality of restriction. Dirichlet minimization is taken over coordinate-isomorphic fibers. □

7 Final-root-enlargement naturality

Theorem 7.1 (Safe-gap atlas invariance)**.** *Let*

$$K_0 \subset S \subset S^+ \subset P \setminus (Q_L \cup Q_R)$$

be compact connected roots, and put $S_w = \phi_w(S)$, $S_w^+ = \phi_w(S^+)$. Then every puzzle piece, contact set, endpoint itinerary, flag, and trace-atlas map is unchanged. The inclusions $S_w \subset S_w^+$ commute with all shift and parent maps; only the internal carrier/cap response can change.

Proof. The atlas depends only on the fixed puzzle and pseudo-puzzle boundaries and on the finite overlay. The safe-gap enlargement does not meet them. The inclusions and commuting squares follow from injectivity of ϕ_w and $g \circ \phi_a = \mathrm{id}$. □

For a certified finite overlay refinement $\mathscr{O} \preceq \mathscr{O}^+$, assume every new child endpoint is an exact pullback or typed source endpoint, every branch is atomwise homeomorphic, and all protected sides are retained. Let π_P, π_a forget the subdivision.

Theorem 7.2 (Refinement naturality)**.** *The atom, flag and trace diagrams commute:*

$$\pi_P p_a^+ = p_a \pi_a, \qquad \pi_P \sigma_a^+ = \sigma_a \pi_a,$$

$$\boxed{\mathrm{Sub}_{Q_a} \mathcal{R}_a = \mathcal{R}_a^+ \,\mathrm{Sub}_P\,.}$$

Introducing and then minimizing the refined internal traces reconstructs the coarse response exactly.

Remark 7.3 (Promotion boundary). This proves naturality of a certified collision-free final overlay. It does not prove that an arbitrary final-root enlargement is collision-free or proof-complete. No-new-demand stability still requires demanded-atom bijection or future observability on every coarse fiber.

8 Recursive response grammar and finite-state observables

Let $\mathcal{B}$ be the exact one-step Dirichlet form on the root terminals and the two child trace interfaces, and let Q_0 be a terminal cap response. With $\star$ denoting exact compatible-trace gluing, define

$$Q_{r+1} = \mathcal{B} \star \left(Q_r^{(L)} \oplus Q_r^{(R)}\right).$$

Theorem 8.1 (Binary response grammar)**.** *Q_r is exactly the response of the normalized depth-r binary assembly. It has a shared recursive circuit of size $O(r)$, although its fully expanded leaf interface may have 2^r ports. The recursion is natural under the normalization and enlargement maps above.*

Proof. The depth-$r+1$ domain is one root cell with one normalized depth-r assembly on each child. Apply Sobolev domain decomposition and induction. Sharing the identical child response symbol gives one new circuit node per depth. □

Let M_L, M_R be $d \times d$ coefficient transitions and R_L, R_R be $w \times w$ branch-feature transports. For $s = a_1 \cdots a_r$, write $M_s = M_{a_1} \cdots M_{a_r}$ and $R_s = R_{a_1} \cdots R_{a_r}$.

Theorem 8.2 (Matrix-product future quotient)**.** *For coefficient vectors u, v and feature vector x,*

$$\sum_{s \in \{L,R\}^r} (u^T M_s v) R_s x = (u^T \otimes I_w)(M_L \otimes R_L + M_R \otimes R_R)^r (v \otimes x).$$

Hence a universal d-state coefficient transducer and w-dimensional branch feature module give an exact future quotient of dimension at most dw, independent of 2^r.

Proof. Expand the r-th power. Its terms are exactly $M_s \otimes R_s$, indexed by binary words, and contract the first tensor factor with u, v. □

Fixed-block KKL Value moment

Let T be one source-licensed finite terminal Hubbard tree, $E = E(T)$, and $\mathcal{S} = E \times E$. Let v, w be the Perron weights and λ the core eigenvalue used by the KKL Value construction. For a level-m arc γ, define

$$z_m(\gamma)_{e,e'} = \sum_{\substack{\alpha:\, f_*^m \alpha = e \\ (\iota^m)_* \alpha = e'}} \langle \alpha, \gamma \rangle, \qquad a_{e,e'} = v(e)\sqrt{w(e')}.$$

For a WAD X, put

$$\mathsf{M}_m(X) = \sum_\gamma X(\gamma) z_m(\gamma) z_m(\gamma)^T.$$

Theorem 8.3 (Exact fixed-block Value state)**.**

$$\boxed{\mathrm{Val}(X) = \lambda^{-2m} \langle a a^T, \mathsf{M}_m(X) \rangle.}$$

Thus one fixed terminal tree has an exact positive homogeneous Value state of dimension at most $|E|^2(|E|^2+1)/2$. If a univalent binary branch induces an incidence matrix P_a on $\mathbb{R}^{\mathcal{S}}$, then

$$\mathsf{M}(b_a^* X) = P_a \mathsf{M}(X) P_a^T,$$

and the joint binary transport is

$$\mathsf{M} \longmapsto P_L \mathsf{M} P_L^T + P_R \mathsf{M} P_R^T.$$

Proof. The KKL coefficient of a pulled bivertical arc α is $\lambda^{-m} v(f_*^m \alpha)\sqrt{w((\iota^m)_* \alpha)}$. Grouping by the pair $(f_*^m \alpha, (\iota^m)_* \alpha)$ gives $\ell_m(\gamma) = \lambda^{-m} a^T z_m(\gamma)$. Substitute this into $\mathrm{Val}(X) = \sum_\gamma X(\gamma) \ell_m(\gamma)^2$. Univalent branch transport preserves intersection numbers, yielding the congruence formula. □

Warning 8.4. The matrix-product theorem and the fixed-block moment theorem do not provide a universal state by themselves. The terminal edge dimension may grow, and a virtual block still needs an actual typed factor to the terminal bivertical incidence. Arbitrary independent leaf coefficients still require exponentially many degrees of freedom.

9 Status and next gate

The following fixed-block binary modules are proved by the preceding arguments: source contact recovery, complete contact recursion, complete cylinder contacts, endpoint-germ itineraries, the one-to-many pseudo-trace atlas, word functoriality, normalized-overlay naturality, safe-gap atlas invariance, and certified-refinement naturality.

The broad direct-decoration and MLC program still requires a complete source Wave horizon, protected final inventory, a universal future-observable response/WAD/Value quotient, MBTT, coercivity and rigidity. The next major analytic theorem should construct a uniformly controlled binary trace/Value future quotient and a strict complete-block transition. No MLC claim is made.